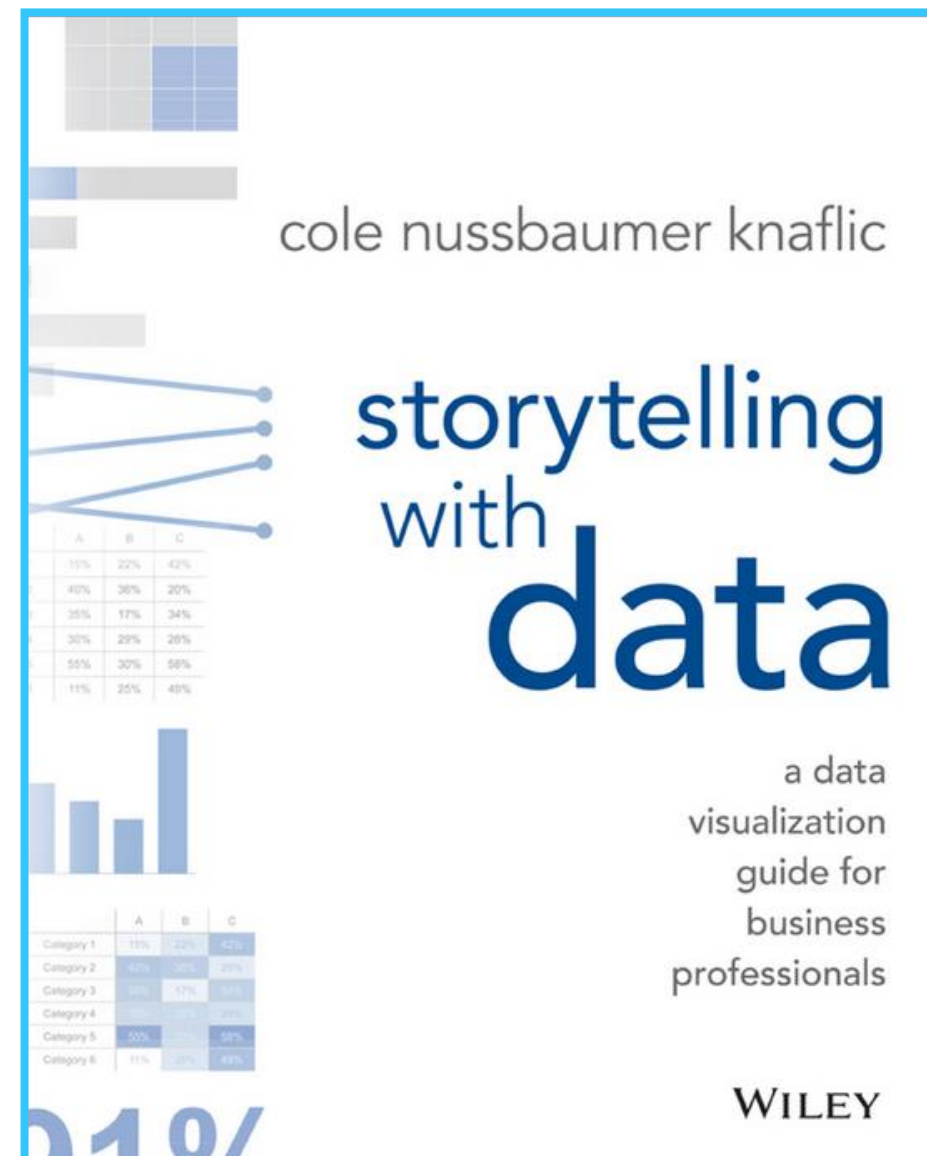


Data Storytelling:

Turning Data into Actionable Insights

by Boris Naydnenov



Make your data make them take action

1. Understand the context + Six Thinking Hats
2. Choose an appropriate visual display
3. Eliminate clutter
4. Focus attention where you want it
5. Think like a designer
6. Tell a story + How Pixar tells successful stories

1. Understand the context

Who is your audience?

Who are you to them?

What do you need them to know or do?

What data is available that will help make my point?

Issue:

Kids have bad attitudes about science

Demonstrate Issue:

show student assignment grades over course of year

Ideas for overcoming issue, including pilot program

Describe pilot program - goals, etc.

Show before & after survey data to demonstrate success of program

RECOMMENDATION:
pilot was a success let's expand it we need \$\$

Six Thinking Hats

DECISION MAKING



Yellow
positivity



Green
creativity



Red
emotions



White
data, rationality



Black
negativity, caution



Blue
process, control

DECISION MAKING

Prioritize by weighing impact against the effort required.

DECISION MAKING

Avoid jumping to conclusions.
Make decisions based on reality.

Make faster decisions with incomplete data.



Cynefin framework

DECISION MAKING

Make sense of different situations to choose an appropriate



Six Thinking Hats

DECISION MAKING

Look at a decision from different perspectives



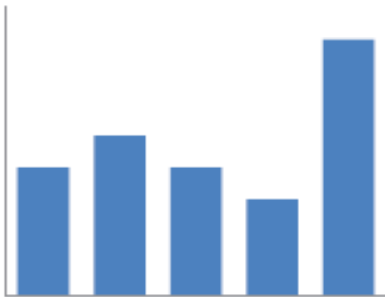
Confidence determines speed vs. quality

DECISION MAKING

Determine a trade-off between

[Untools.co](https://untools.co) - cool website

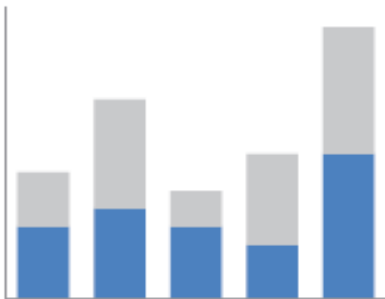
2. Choose an appropriate visual display



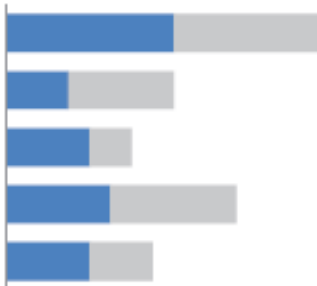
Vertical bar



Horizontal bar



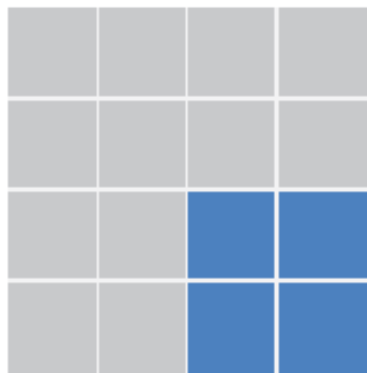
Stacked vertical bar



Stacked horizontal bar



Waterfall



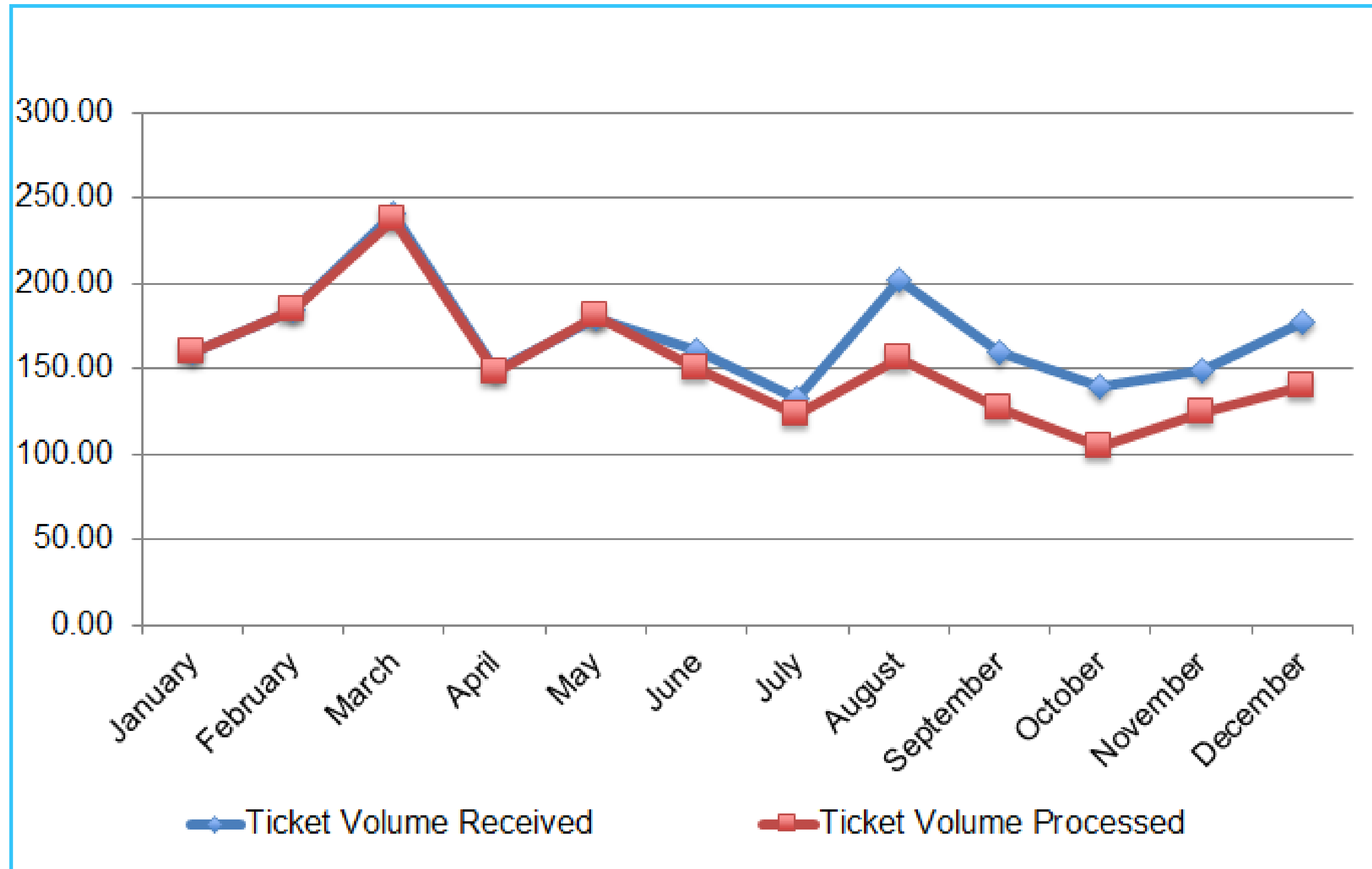
Square area



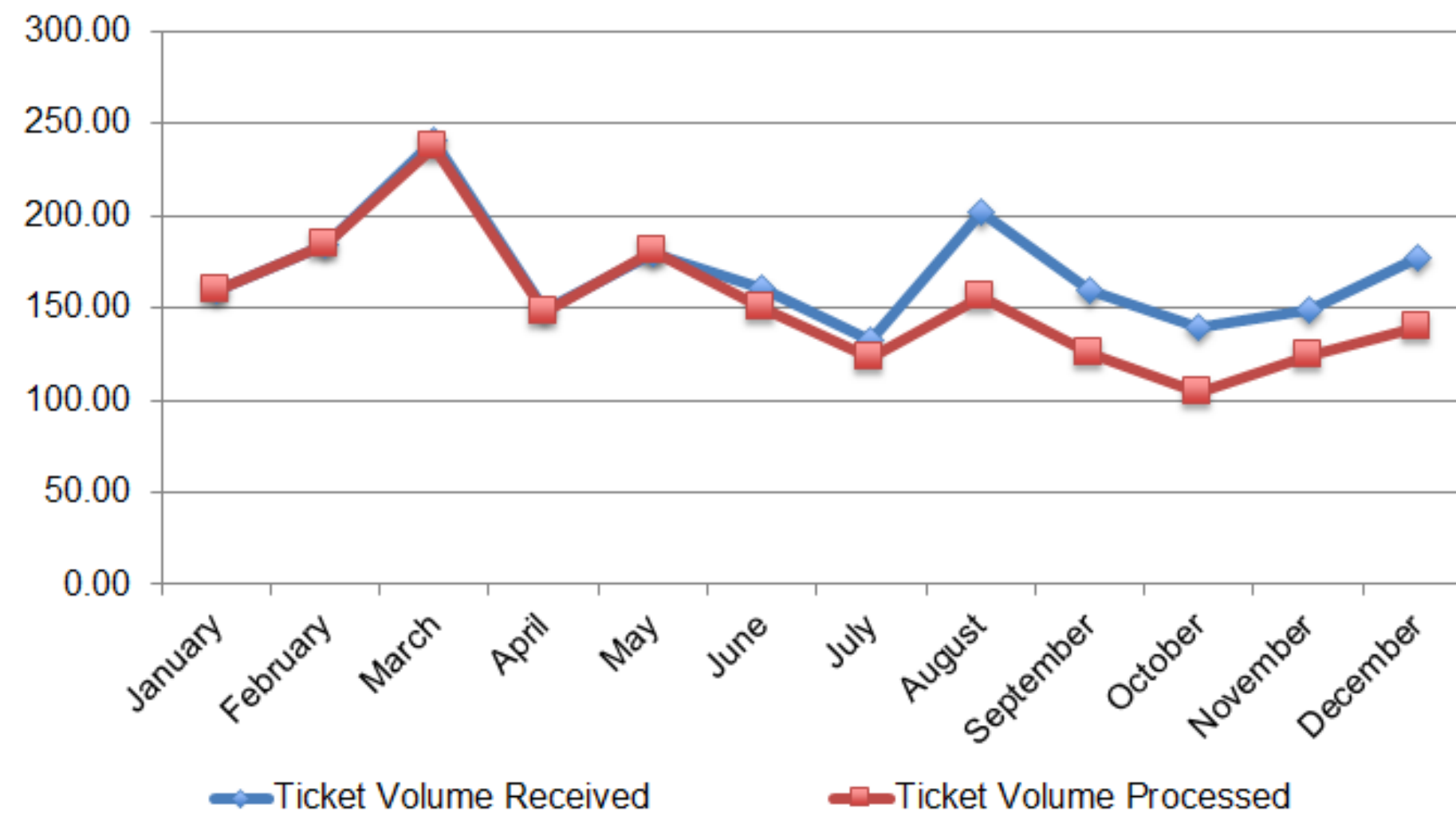
3. Eliminate clutter

Decluttering: step-by-step

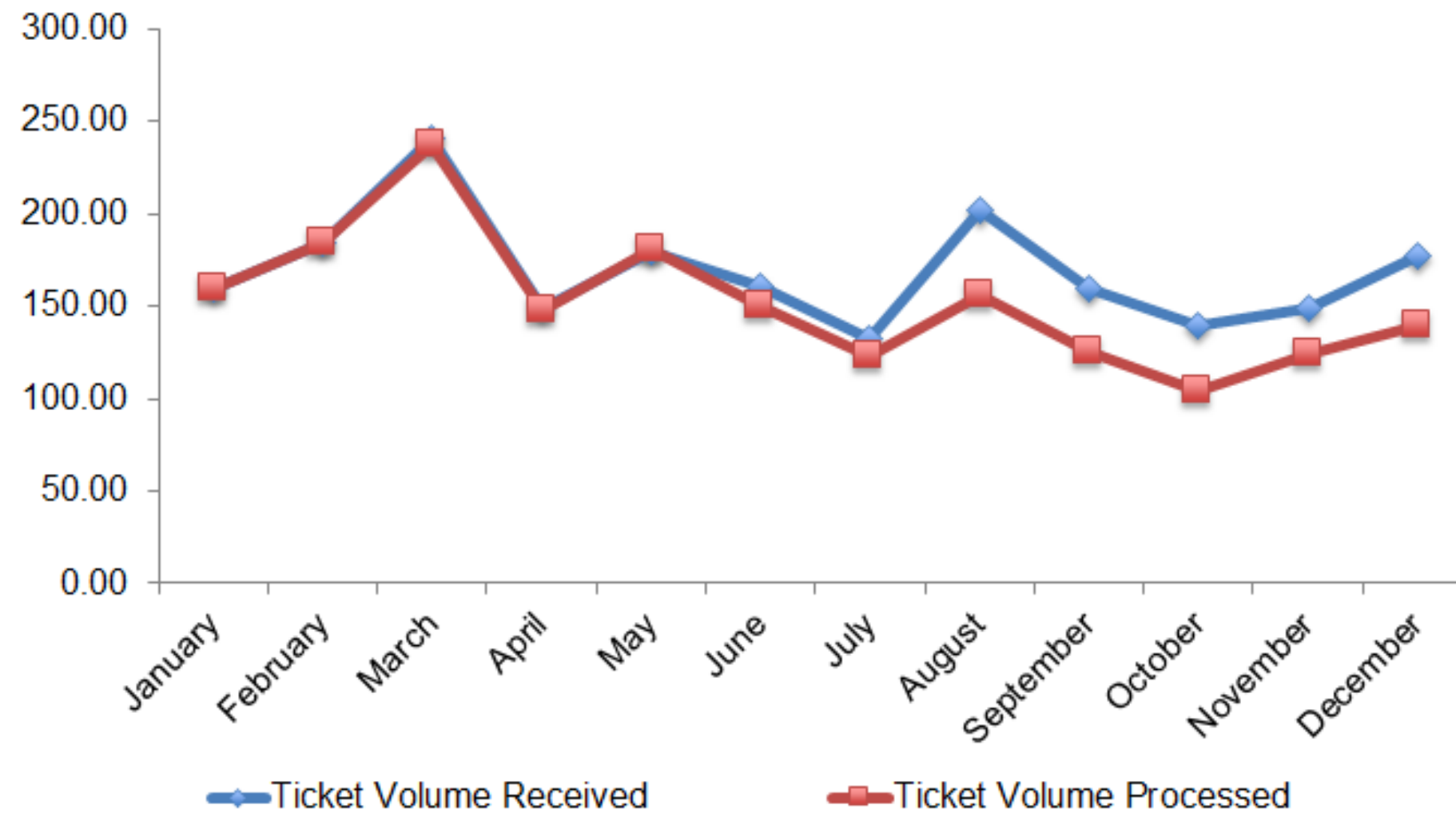
1. Remove chart border
2. Remove gridlines
3. Remove data markers
4. Clean up axis labels
5. Label data directly
6. Leverage consistent color



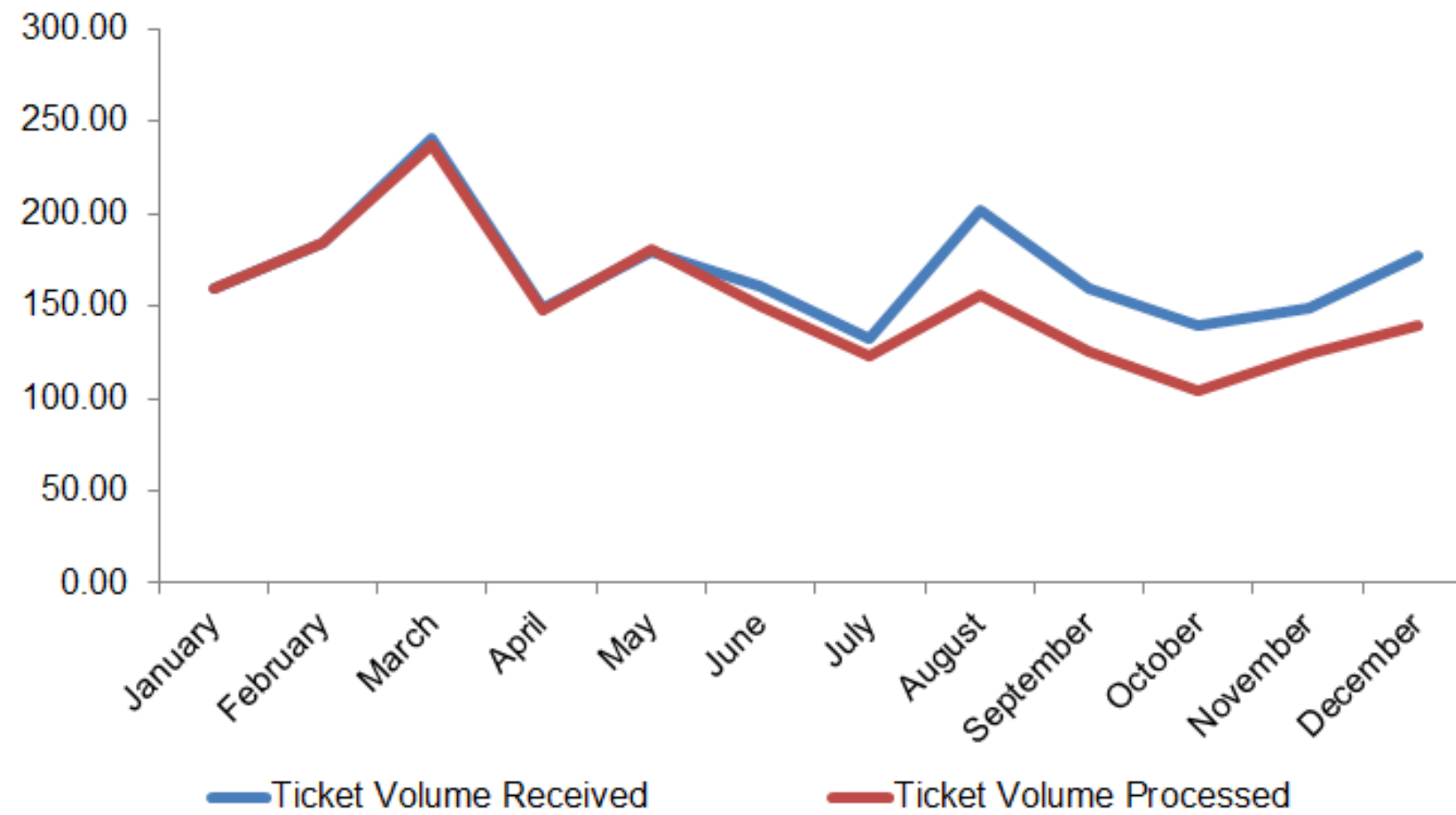
Remove chart border



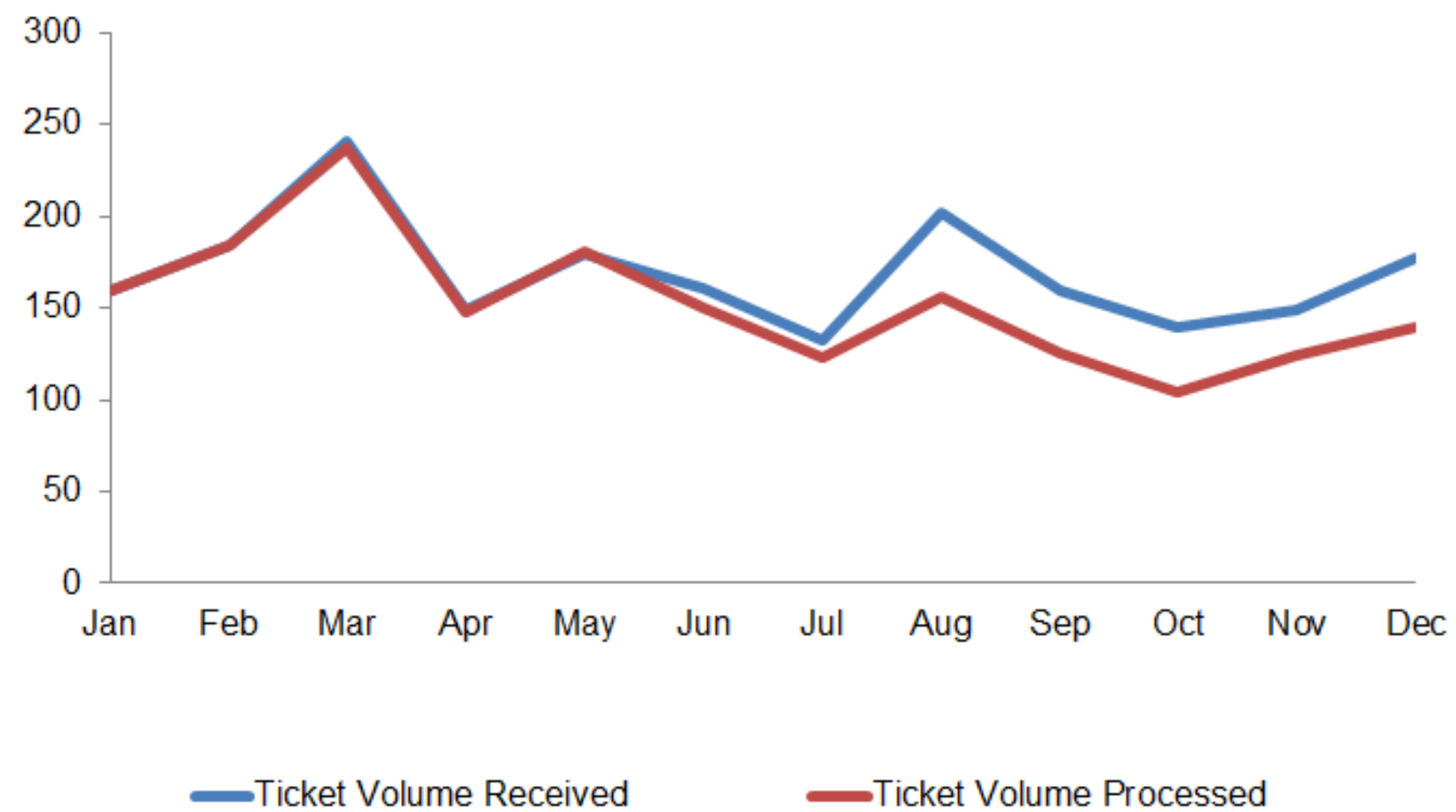
Remove
gridlines



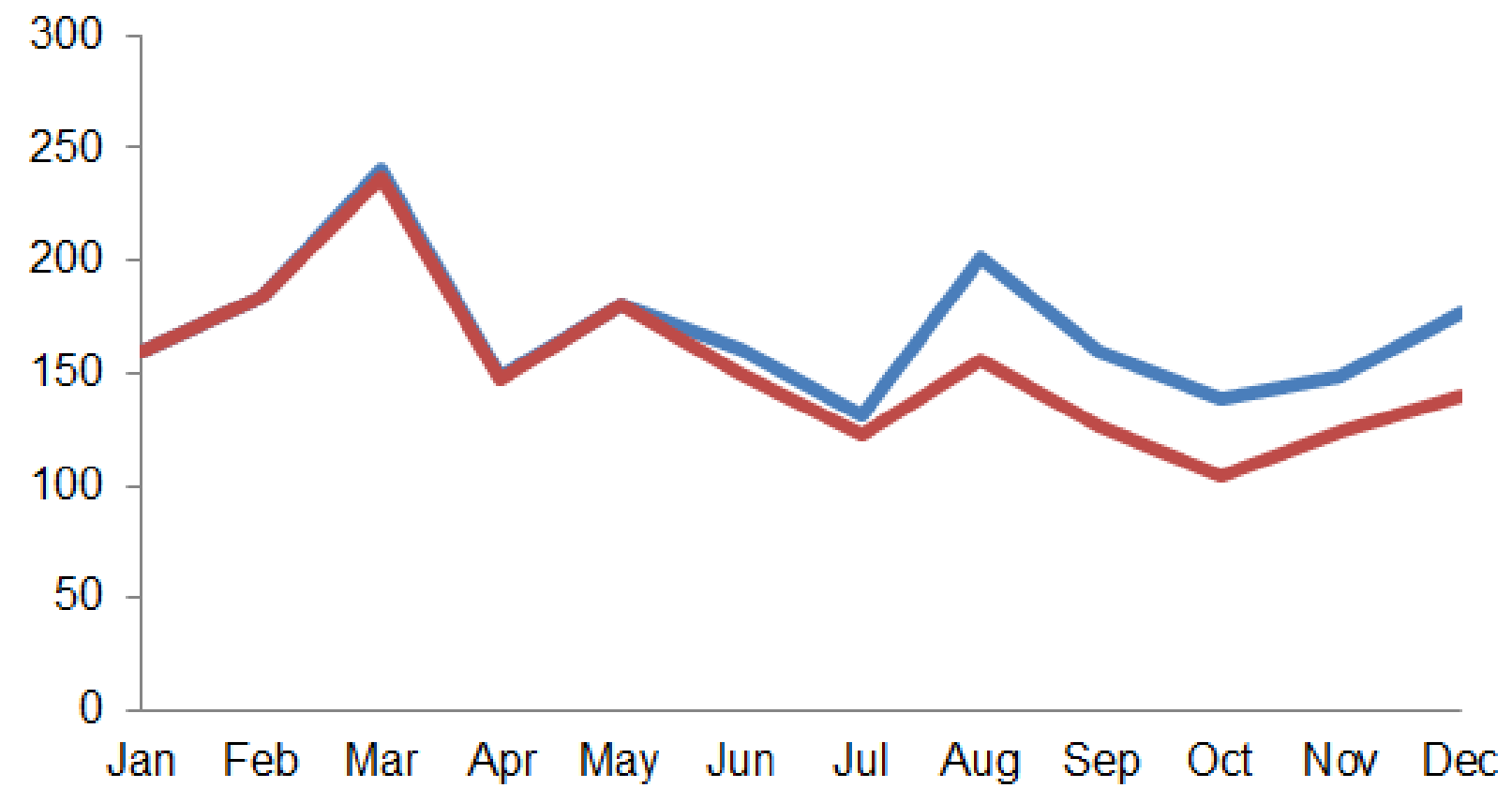
Remove data markers



Clean up axis labels

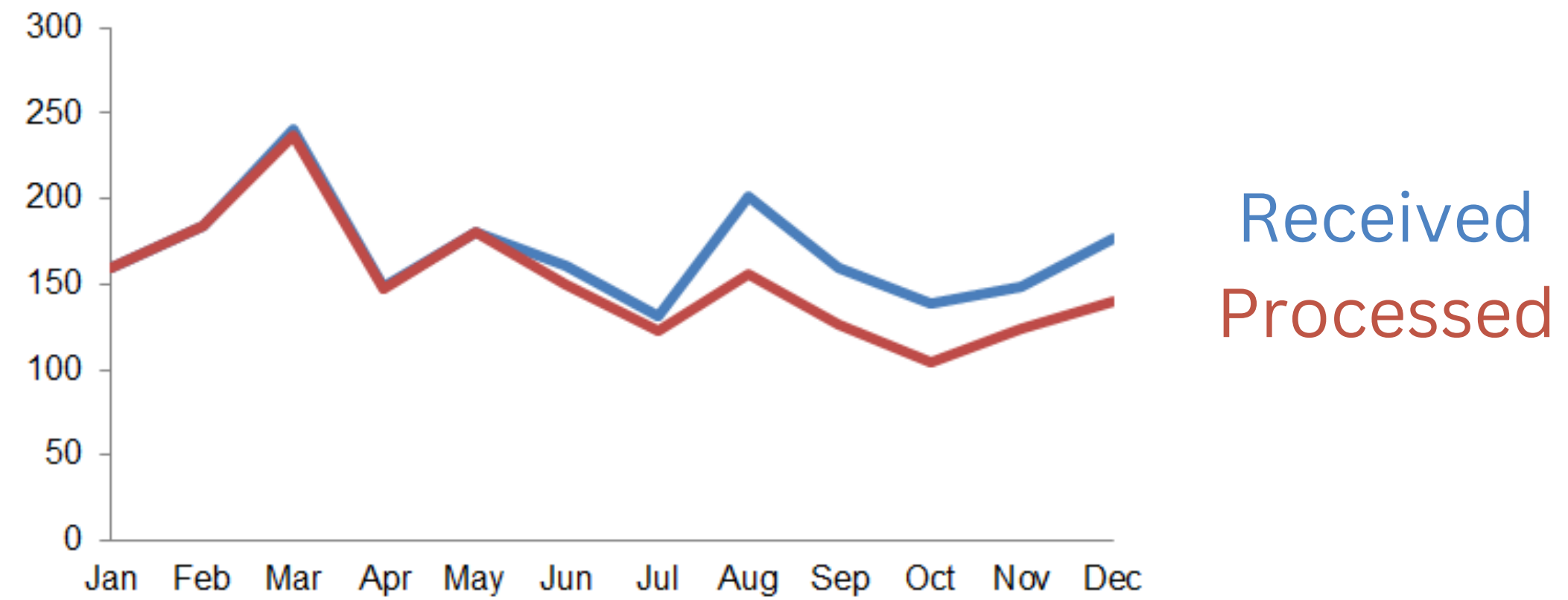


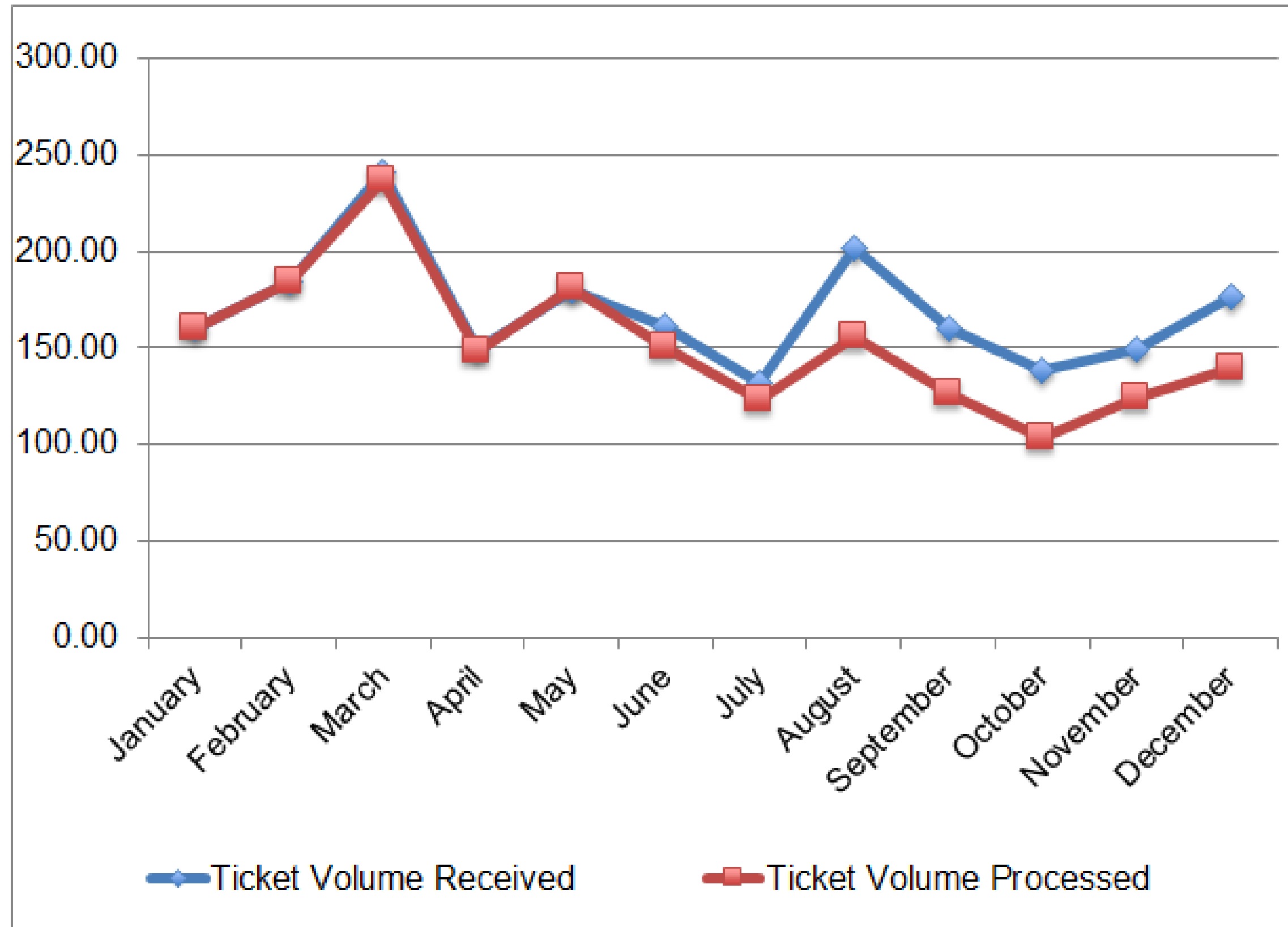
Label data
directly



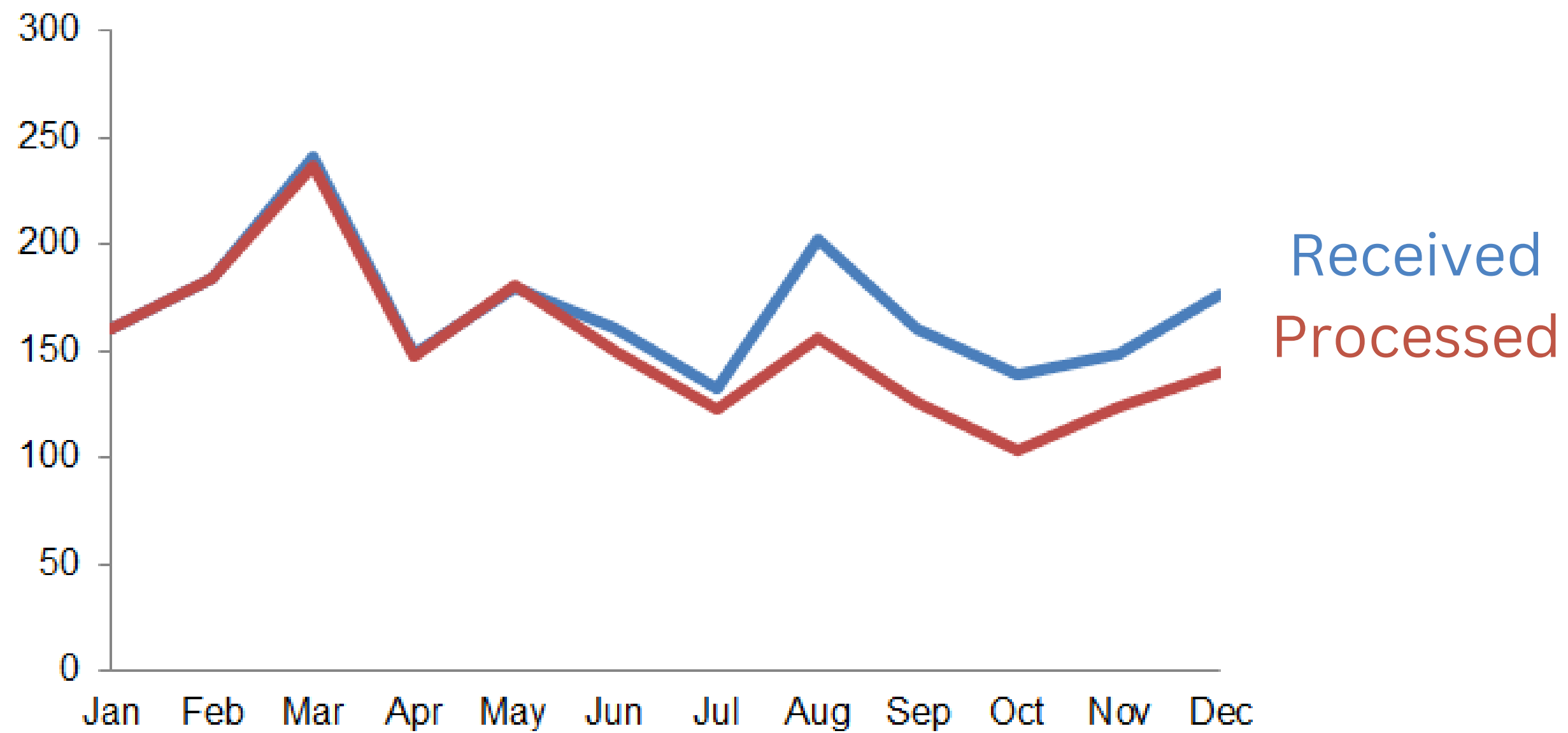
Received
Processed

Leverage consistent color



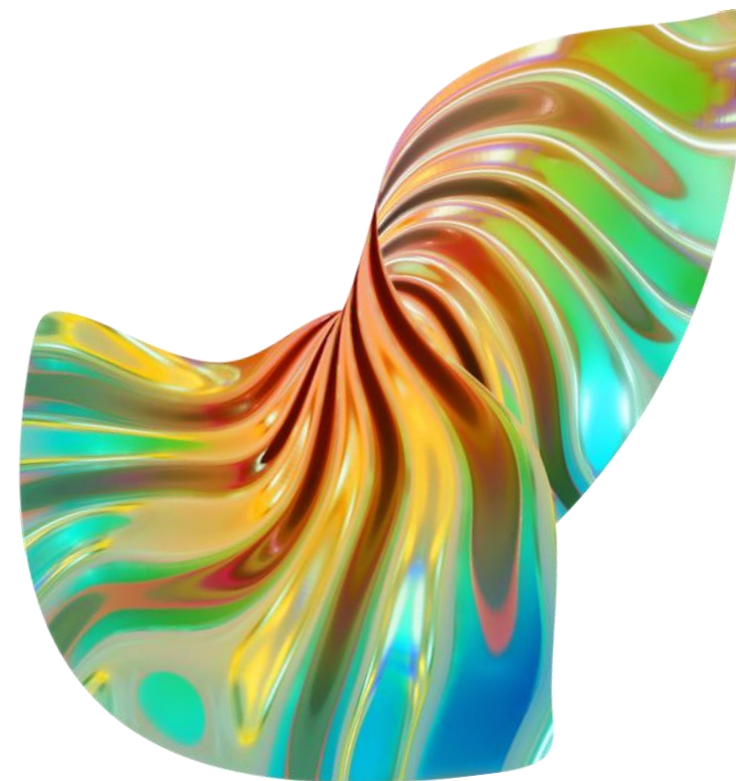


Before

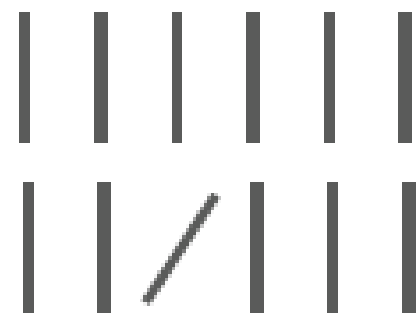


After

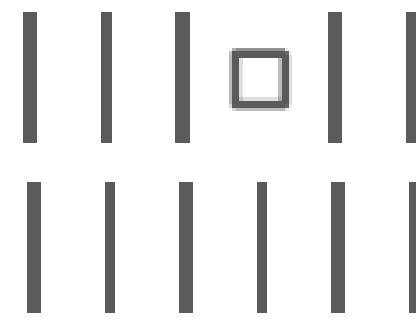
4. Focus attention where you want it



Preattentive attributes signal where to look



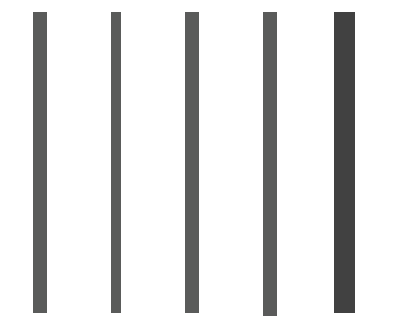
Orientation



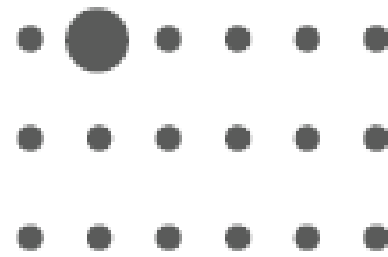
Shape



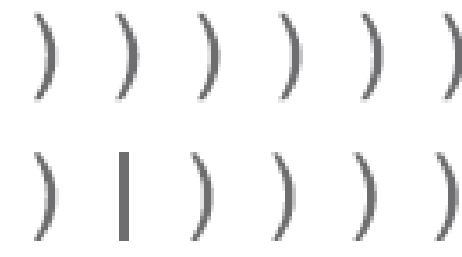
Line length



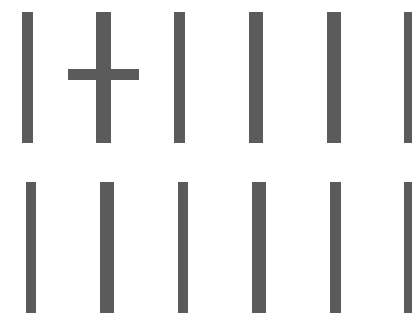
Line width



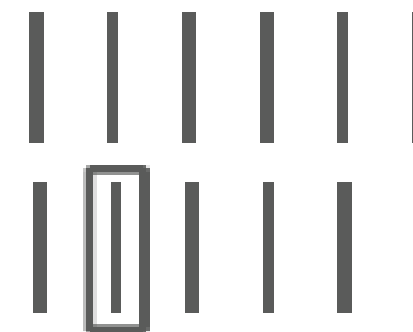
Size



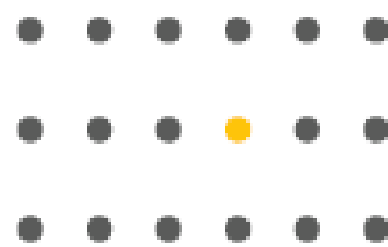
Curvature



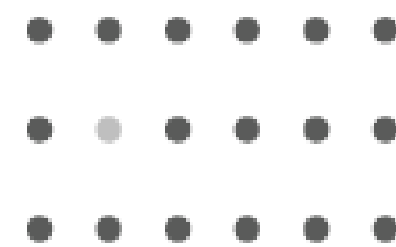
Added marks



Enclosure



Hue



Intensity



Spatial position



Motion

No preattentive attributes

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

Color

What are we doing well? Great Products. **These products are clearly the best in their class.** Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

Size

What are we doing well? Great Products. These products are the best in their class. Replacement parts are shipped when needed. You sent gaskets **without me having to ask**. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

Outline (enclosure)

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The **account manager even called to check in** after normal business hours. You have a great company – keep up the good work!

Bold

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

Italics

What are we doing well? Great Products. These products are clearly the best in their class. *Replacement parts are shipped when needed.* You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

Separate spatially

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask.

Problems are resolved promptly.

Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

Underline (added marks)

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

5. Think like a designer

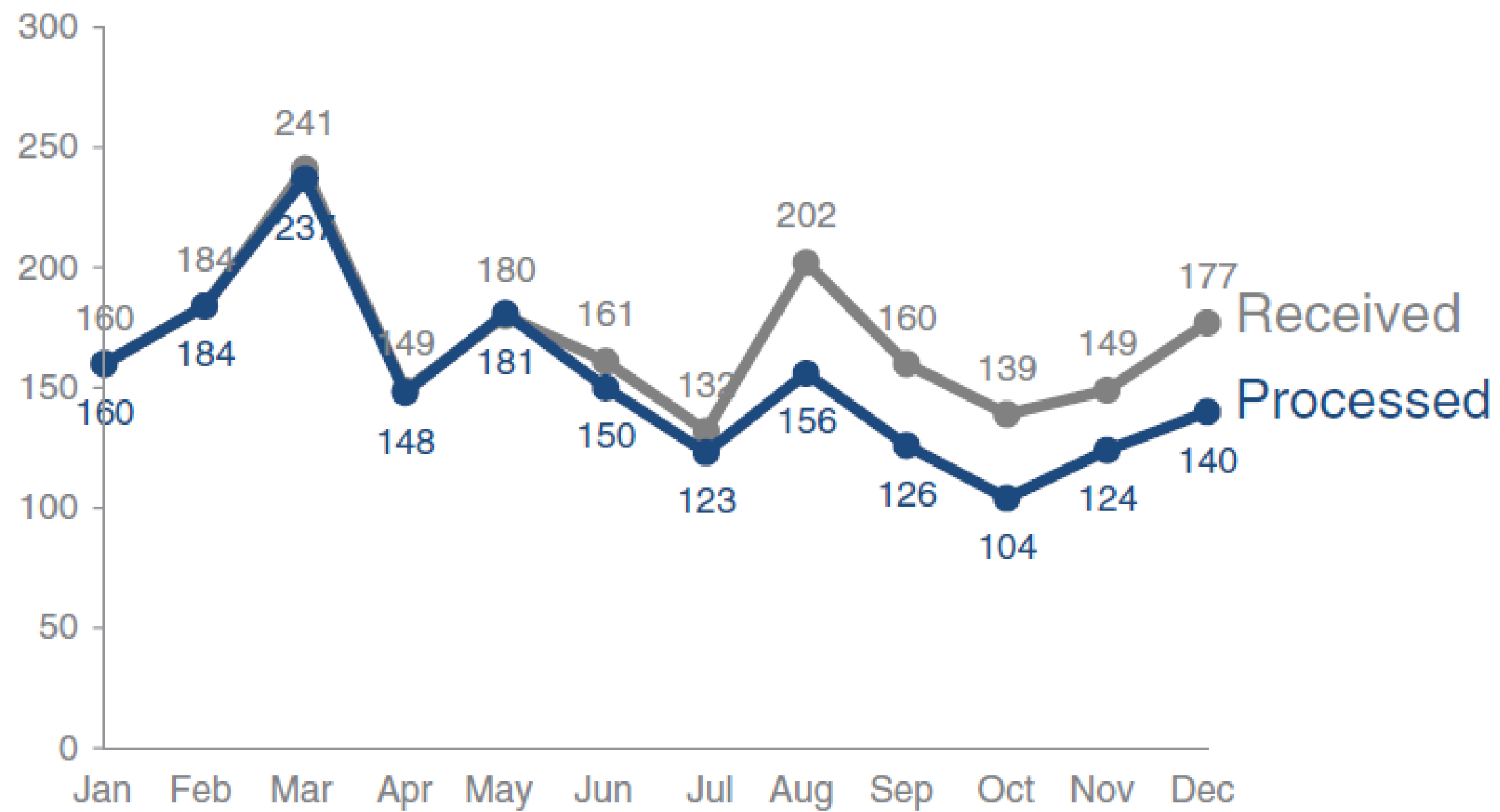
Highlight the important stuff

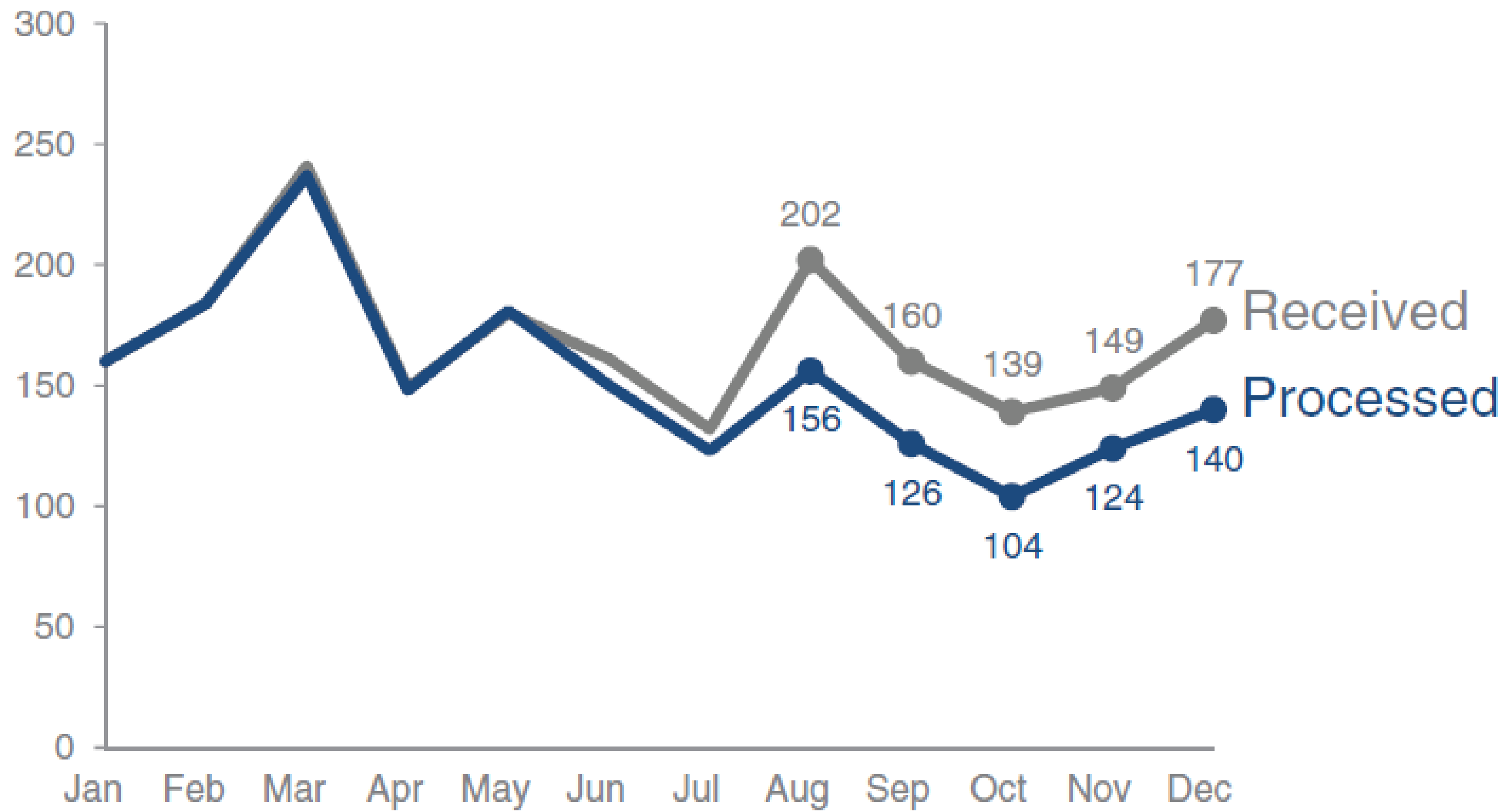
Eliminate distractions

Create a clear visual hierarchy of information

**When you highlight one point in your story,
it can make other points harder to see**

•





Successful

storytelling

1. Simple



2. Purpose

3. Relatable

4. Universal

5. Surprising

6. Emotional

7. Clear structure



keep it
SIMPLE

Successful Storytelling

1. Simple
2. Purpose
3. Relatable
4. Universal
5. Surprising
6. Emotional
7. Clear structure



*Thank
you!*